PRIME FACTOR STUDIO — IN-BOX USER GUIDE

Kinesthetic Number Theory, Proportional Logarithmic Stacking & 12-Socket Lid Tray

CARD 05 / 10 • PRIME-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · 4·Fractions · **5·Primes** · 6·Algebra · 7·Probability · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

FEEL THE PRIMES. MULTIPLY WITH HEIGHT.

Standard base-10 counting blocks add lengths (1+1=2).
Prime Factor blocks are logarithmically proportional:
log(a × b) = log(a) + log(b)
Stacking blocks in a tower multiplies their numbers!

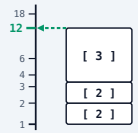
CALIBRATED LOGARITHMIC DATUM

- Unit Datum: Block [2] = 1U = 7.00 mm visible height
- Log Modulus: K_log = 23.2535 mm / decade
- Prime Formula: h_vis(p) = K_log × log10(p)
- Connector Spec: 1.85mm base plug & stacking socket

12-SOCKET LID TRAY & LOGARITHMIC SCALE

Flip 13U lid over to use 12-socket comparison stage.
Equal heights prove equal mathematical products!

VISUAL PROOF: TOWER 12 = [2] + [2] + [3]



▲ SUMMIT = 12
2 × 2 × 3 = 12
Visible: 25.09 mm

Commutative:
[3]+[2]+[2] = same height!

2. KIT PARTS INVENTORY (72 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Prime Block [2]	29x	7.00 mm (1.00U)	Blue
Prime Block [3]	17x	11.09 mm (1.58U)	Gold
Prime Block [5]	8x	16.25 mm (2.32U)	Classic White
Prime Block [7]	4x	19.65 mm (2.81U)	Slate Gray
Prime Block [11]	3x	24.21 mm (3.46U)	Gold
Prime Block [13]	3x	25.90 mm (3.70U)	Classic White
Prime Block [17]	2x	28.61 mm (4.09U)	Slate Gray
Prime Block [19]	2x	29.73 mm (4.25U)	Gold
Prime Block [23]	2x	31.67 mm (4.52U)	Slate Gray
13U Storage Bin	1x	91.0 mm (13U)	84.0mm Waterline
Stage Lid	1x	10.4 mm Stage	12-Socket Lid Tray

TOTAL: 70 Prime Blocks + Storage Bin + Comparison Stage Lid = 72 Parts

4-COLOR HIGH-CONTRAST PALETTE

- 4-Color Palette: Neighboring prime columns have distinct hues.
- Instant Sorting: Enables fast visual sorting across 70 blocks.
- High Durability: Scratch-resistant debossed numerals.
- Tactile Alignment: Connectors mate smoothly without binding.

3. WHY KEEP THIS REFERENCE CARD?

✓ NO LOST PIECE LEFT BEHIND

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ 10-SECOND SELF-CHECKING PACK-UP

Pack-up is a built-in math game! Each floor socket is labeled with its tower's product: 2^{12}=4096, $3^7 \cdot 2$=4374, 5^5=3125, $7^4 \cdot 2$=4802... When all 70 blocks are stacked in their bays, all 12 towers level at the 84.0mm waterline. Stage lid latches firmly for quiet carry.

✓ HANDS-ON TUTORIAL ON REVERSE

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

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FUNDAMENTAL THEOREM OF ARITHMETIC

Every integer > 1 has a unique prime factorization decomposition.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

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SERIES 1 • CARD 05 OF 10 • DOMAIN: FUNDAMENTAL THEOREM OF ARITHMETIC • LOGARITHMIC PROPORTIONAL TOWERS • FLUSH DATUM

LESSON GUIDE: HOW TO FIND THE LEAST COMMON MULTIPLE (LCM)

Step-by-Step Hands-On Tutorial: Finding LCM(12, 18) = 36 & The Law of Conservation of Primes

Grades 4–9 Math & Pre-Algebra • Hands-On Classroom & Home STEM Studio Series

CARD 05 / 10 • PRIME-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

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Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.6.NS.B.4 (LCM & GCD)
- CCSS.MATH.CONTENT.4.OA.B.4 (Factor Pairs)
- TEKS 111.26.B.6.7.A (Prime Factorization)

Grades 4–9 Math & Pre-Algebra

MATHEMATICAL INQUIRY

Which prime blocks are shared by both towers, and which blocks are unique to one tower?

1 STEP 1: FACTOR & BUILD TOWERS 12 & 18

Decompose numbers 12 and 18 into prime building blocks:
- **Tower 12: Stack [2] + [2] + [3] (2 × 2 × 3 = 12)**
- **Tower 18: Stack [2] + [3] + [3] (2 × 3 × 3 = 18)**

Seat each tower in the 12-socket lid tray. Their summits **measure 25.09 mm (Tower 12) and 29.18 mm (Tower 18)!**

2 STEP 2: FIND SHARED OVERLAP (THE GCD)

Stand Tower 12 and Tower 18 side-by-side.
Identify the blocks they share in common (intersection):
• Tower 12 has: [2], [2], [3] | Tower 18 has: [2], [3], [3]
Shared Overlap: Exactly one [2] and one [3]!
Assemble GCD Tower: [2] + [3] = 6 (Summit: 18.09 mm).

3–4 STEPS 3 & 4: ASSEMBLE THE LCM TOWER

3. TAKE PRIME UNION: Take highest power of each prime: two [2]s, two [3]s.
4. STACK LCM TOWER: [2] + [2] + [3] + [3] = 36 (Summit: 36.19 mm)!
Calculator Verification: Press Factor Tree Icon ([PRIME] / [g][FACT]) and run 12 [ENTER] 18 [GCF] 12 [ENTER] 18 [f] [LCM] → 36.

5 CORE PRINCIPLE: CONSERVATION OF PRIMES: A × B = GCD × LCM

Multiply Tower 12 and Tower 18: you have three [2]s and three [3]s. Look at your GCD Tower ([2], [3]) and your LCM Tower ($[2]^2$, $[3]^2$): together they contain the EXACT same prime blocks!
Therefore: A × B = GCD(A, B) × LCM(A, B) = 6 × 36 = 216!

★ EXTENSION INVESTIGATION: FIND LCM(8, 12)

Build Tower 8: [2]+[2]+[2] and Tower 12: [2]+[2]+[3]. Prime union takes three [2]s and one [3]
→ LCM = 24 (28.09mm)! (Scan QR for 3D animated solution!)

→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

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